

MATH – 241
Engineering Mathematics I
Problem Set – 8

Question 1. For this question, you are expected to implement curve fitting on MATLAB. The objective of curve fitting is to minimize below cost function by choosing $\underline{\theta}$ for given (x_k, y_k) where $1 \leq k \leq m$:

$$J(\underline{\theta}) = \sum_{k=1}^m (y_k - f(x_k; \underline{\theta}))^2$$

where the model function $f(x; \underline{\theta})$ is defined as:

$$f(x; \underline{\theta}) = \sum_{i=1}^n g_i(x) \theta_i \text{ and } g_i(x) = e^{-\frac{(x-i\mu)^2}{2\sigma^2}}$$

- (a) For $\underline{y} = [y_1, y_2, \dots, y_m]^T$ and $\underline{\theta} = [\theta_1, \theta_2, \dots, \theta_n]^T$, formulate the problem to reduce it to the minimization of

$$\min_{\underline{\theta}} \|\underline{y} - \underline{A} \underline{\theta}\|^2.$$

Identify the elements of $\underline{A}$ and obtain a closed form solution for optimal $\underline{\theta}^*$.

- (b) For $\underline{y}$ and $\underline{x}$ given in the attached file and the parameters as $\mu = 3$, $\sigma^2 = 4$, $m = 22$, and $n = 7$ obtain the best fit parameters $\underline{\theta}^*$ and plot:

$$\text{plot}(\underline{x}, \underline{y}, 'o', \underline{x}, \underline{y}_s, '-')$$

where $\underline{y}_s = [y_s(1), y_s(2), \dots, y_s(22)]^T$, and $y_s(k) = f(x_k; \underline{\theta}^*)$.

Question 2. Does there exists a matrix A such that the entire family of $A + cI$ is invertible for all complex numbers c ? Find a real matrix with $A + rI$ invertible for all real r .

Question 3. If K is a skew-symmetric matrix, show that $Q = (I - K)(I + K)^{-1}$ is a orthogonal matrix. Find Q if $K = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$.

Question 4. Prove

(a) If $\lambda \neq 0$ is an eigenvalue of A , $\frac{1}{\lambda}|A|$ is an eigenvalue of $\text{adj}(A)$.

(b) If v is an eigenvector of A , then v is also an eigenvector of $\text{adj}(A)$.

Question 5. If $Ax = \lambda_1 x$ and $A^T y = \lambda_2 y$ (all real), show that $x^T y = 0$.

Question 6. For the matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 1 & 2 \end{bmatrix}$, find $f(A)$ where $f(x) = \frac{e^x + e^{-x}}{e^x - e^{-x}}$